

NOARLUNGA, Australia

WMO: 948080

Lat: 35.1586S

Lon: 138.5058E

Elev: 180

StdP: 14.60

Time Zone: 9.50 (AUA)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.9	44.4	24.8	19.0	75.9	28.2	22.3	73.7	26.4	57.6	24.0	55.5	5.1	60	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.3	97.9	64.7	93.6	63.9	89.7	62.7	68.9	84.7	67.3	83.5	65.8	82.2	10.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	91.8	73.2	62.1	83.9	72.1	59.9	77.6	71.1	33.1	84.4	31.8	83.7	30.6	82.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.7	22.8	20.1	39.6	106.0	1.1	3.5	38.8	108.5	38.1	110.6	37.5	112.5	36.7	115.1
			37.0	72.0	1.4	1.9	36.0	73.3	35.1	74.4	34.3	75.5	33.3	76.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.3	72.1	71.3	68.6	64.4	58.6	54.0	53.1	53.8	57.1	61.0	65.5	68.7	(6)
	DBStd	9.16	8.06	7.60	7.07	5.82	4.25	3.30	3.30	4.15	5.33	6.84	7.95	8.19	(7)
	HDD50	21	0	0	0	0	0	5	8	7	1	0	0	0	(8)
	HDD65	1942	10	14	33	79	205	331	370	348	248	170	89	45	(9)
	CDD50	4510	685	595	576	432	268	125	104	125	214	341	466	579	(10)
	CDD65	953	231	189	144	61	7	0	0	1	11	45	105	159	(11)
	CDH74	9890	2572	1939	1375	424	17	0	0	3	71	492	1156	1841	(12)
	CDH80	4486	1336	901	584	114	1	0	0	7	151	486	906		(13)
(14) Wind	WSAvg	10.0	11.6	11.3	10.3	9.2	8.6	8.3	9.8	9.6	9.7	10.5	10.8	10.8	(14)
(15) Precipitation	PrecAvg	20.2	0.7	0.7	0.9	1.7	2.6	2.6	3.0	2.6	2.0	1.7	1.0	0.9	(15)
	PrecMax	29.4	2.3	5.6	3.4	5.0	5.6	5.3	7.8	4.2	4.8	4.7	4.4	2.1	(16)
	PrecMin	12.2	0.0	0.0	0.0	0.1	0.5	0.6	0.8	0.7	0.6	0.2	0.1	0.0	(17)
	PrecStd	4.2	0.6	1.1	0.9	1.3	1.3	1.2	1.3	1.0	1.0	1.1	0.7	0.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.5	100.6	97.5	89.1	75.9	67.8	66.4	72.8	80.6	89.7	96.1	101.2	(19)
		MCWB	66.4	67.3	63.6	62.0	56.9	55.3	52.7	54.3	56.6	60.0	63.1	65.0	(20)
	2%	DB	98.7	94.3	90.9	82.8	72.1	63.5	62.6	67.8	75.8	84.0	90.4	95.6	(21)
		MCWB	65.4	65.5	63.0	60.0	56.3	53.7	51.3	52.6	55.4	58.2	61.4	63.2	(22)
	5%	DB	93.1	90.1	86.7	78.7	68.4	61.1	60.2	64.2	71.7	79.5	85.6	89.9	(23)
		MCWB	64.2	64.0	63.1	58.7	55.8	53.3	51.3	51.9	54.3	56.8	60.4	62.0	(24)
	10%	DB	87.3	85.3	81.9	74.9	65.5	59.4	58.3	61.1	67.7	74.8	81.0	84.4	(25)
		MCWB	63.2	63.3	61.8	57.6	55.3	52.6	51.3	51.1	52.8	55.9	59.7	60.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.1	71.1	69.1	65.0	61.0	57.9	56.3	56.9	59.3	63.3	67.1	69.3	(27)
		MCDB	83.5	89.4	82.2	75.1	67.6	62.1	61.6	65.0	72.9	79.4	82.4	82.4	(28)
	2%	WB	69.0	68.4	67.1	63.2	59.4	56.1	54.7	55.0	57.5	60.8	64.7	66.7	(29)
		MCDB	86.5	83.9	80.5	73.5	65.8	60.5	58.8	62.3	69.0	76.2	81.1	82.9	(30)
	5%	WB	67.3	66.7	65.4	61.7	58.1	54.9	53.6	53.8	56.0	59.1	63.3	64.7	(31)
		MCDB	85.8	82.6	79.8	72.3	64.5	59.4	57.4	60.5	66.5	72.9	78.7	82.4	(32)
	10%	WB	65.3	65.1	63.5	60.1	56.8	53.8	52.7	52.7	54.7	57.5	61.3	62.9	(33)
		MCDB	83.4	81.9	78.7	71.2	63.4	58.3	56.7	58.8	64.2	70.4	77.3	80.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	17.7	16.3	14.1	10.7	9.7	9.2	11.1	13.5	16.5	18.0	18.4	(35)
		MCDBR	27.4	24.2	22.7	19.5	13.6	11.2	10.6	15.7	20.8	25.3	25.5	27.1	(36)
	5% WB	MCWBR	9.3	9.0	8.4	8.3	7.1	7.0	6.5	7.8	8.7	9.5	9.1	9.9	(37)
		MCWBR	22.1	20.5	19.1	15.0	10.5	9.2	8.5	12.1	16.7	21.0	22.1	21.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.342	0.342	0.323	0.317	0.308	0.300	0.299	0.310	0.332	0.343	0.345	0.338	(40)
		taud	2.549	2.565	2.612	2.613	2.618	2.620	2.616	2.569	2.502	2.477	2.501	2.546	(41)
	Ebn at Noon		315	309	303	287	272	266	272	285	295	305	312	318	(42)
		Edh at Noon	34	32	29	25	22	21	22	26	32	35	36	35	(43)
(44) All-Sky Solar Radiation	RadAvg		2446	2132	1731	1253	855	704	765	1055	1452	1888	2235	2404	(44)
	RadStd		123	104	77	70	36	45	35	63	110	115	97	95	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-176	+290	+126

Nomenclature: See separate page